

Constancy of the Wong Matrix for Three-Dimensional Finite-Dimensional Estimation Algebras

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Abstract

For a three-dimensional nonlinear filtering system with smooth drift and observation functions and orthogonal diffusion, we prove that every finite-dimensional estimation algebra of linear rank two and quadratic rank zero has a constant Wong matrix. The proof remains entirely in the C^∞ category. Starting from the published function-element theorem and affine-structure theorems, we first give a self-contained all-order proof of constancy of the visible entry ω_{12} : a quotient of normally ordered principal symbols and triangular growth at every order exclude all nonzero visible slopes. We then divide the remaining parameters into three mutually exclusive sectors. When a hidden slope is nonzero, a smooth Euler finite-module lemma shows that the normally ordered coefficients of a finite-dimensional adjoint module are automatically polynomial in the hidden Euler variable, and that every weight component is obtained by a genuine Lie word projection furnished by the Chinese remainder theorem. This yields ∂_3^2 and then a strictly increasing differential-order Lie ladder. When the hidden slopes vanish but the visible-slope matrix is nonzero, a rigidity theorem for the potential followed by a weighted associated-graded argument excludes hidden profiles of every degree. The argument covers the rank-one boundary, arbitrary constant tails, and every zero-parameter branch; it uses neither analyticity, a maximal-rank hypothesis, nor extrapolation from finitely many orders.

Keywords: Nonlinear filtering; finite-dimensional estimation algebra; Wong matrix; non-maximal rank; Euler operator.

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This paper settles the remaining quadratic-rank-zero branch of the original classification. The precise result is as follows.

Theorem 0.1 (Main theorem). *Consider*

$$dx(t) = f(x(t)) dt + g(x(t)) dv(t), \quad x(t) \in \mathbb{R}^3, \quad (0.1)$$

$$dy(t) = h(x(t)) dt + dw(t), \quad (0.2)$$

where $f \in C^\infty(\mathbb{R}^3; \mathbb{R}^3)$, $h \in C^\infty(\mathbb{R}^3; \mathbb{R}^m)$, and $g: \mathbb{R}^3 \rightarrow O(3)$. Suppose that the associated estimation algebra E is finite-dimensional, the state dimension is three, the linear rank is two, and the quadratic rank is zero. Then its Wong matrix Ω is constant.

More explicitly, in the affine normal form used below,

$$\omega_{12} = b_1x_1 + b_2x_2 + b_0, \quad (0.3)$$

$$\omega_{13} = k_1x_1 + k_2x_2 + k_3x_3 + k_0, \quad (0.4)$$

$$\omega_{23} = h_1x_1 + h_2x_2 + h_3x_3 + h_0, \quad (0.5)$$

one necessarily has

$$b_1 = b_2 = k_1 = k_2 = k_3 = h_1 = h_2 = h_3 = 0.$$

Consequently,

$$\Omega = \begin{pmatrix} 0 & b_0 & k_0 \\ -b_0 & 0 & h_0 \\ -k_0 & -h_0 & 0 \end{pmatrix}.$$

1 Notations and Prelimeries

1.1 Notations

To avoid confusion with the affine Wong coefficients h_0, h_1, h_2, h_3 below, write the components of the observation map as $h = (q_1, \dots, q_m)$. Set

$$D_i = \partial_i - f_i, \quad \eta = \sum_{i=1}^3 (\partial_i f_i + f_i^2) + \sum_{j=1}^m q_j^2, \quad L_0 = \frac{1}{2} \sum_{i=1}^3 D_i^2 - \frac{1}{2} \eta. \quad (1.1)$$

On expansion,

$$L_0 = \frac{1}{2} \Delta - \sum_i f_i \partial_i - \sum_i \partial_i f_i - \frac{1}{2} \sum_j q_j^2.$$

The estimation algebra is

$$E = \text{Lie}_{\mathbb{R}} \langle L_0, q_1, \dots, q_m \rangle.$$

Functions are identified with multiplication operators. Following the definition in the literature, let

$$\mathcal{L}(E) = E \cap \{\text{homogeneous linear polynomials}\};$$

the linear rank is $\dim_{\mathbb{R}} \mathcal{L}(E)$. The assertion that the quadratic rank is zero means here that E contains no nonzero polynomial multiplication operator of degree exactly two. These are precisely the two rank assumptions in the main theorem; they include neither maximal rank, analyticity, nor any additional elimination condition. The Wong entries are defined by

$$\omega_{ij} = [D_j, D_i] = \partial_i f_j - \partial_j f_i. \quad (1.2)$$

The definitions of the estimation algebra, linear rank, and Wong matrix agree with Jiao [4, Definitions 2.3–2.5, pp. 21–22]; quadratic rank zero is understood according to the convention in the preceding sentence.

Let $\mathcal{U}_r$ denote the space of smooth-coefficient differential operators of order at most r . Normal ordering takes the form

$$P = \sum_{\alpha} a_{\alpha}(x) \partial^{\alpha},$$

and

$$(a \partial^{\alpha})(b \partial^{\beta}) = \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} a(\partial^{\gamma} b) \partial^{\alpha - \gamma + \beta}. \quad (1.3)$$

In particular,

$$[D_i, u] = \partial_i u, \quad [L_0, u] = \sum_i (\partial_i u) D_i + \frac{1}{2} \Delta u, \quad (1.4)$$

and

$$[L_0, D_i] = \sum_j \omega_{ij} D_j + \frac{1}{2} \sum_j \partial_j \omega_{ij} + \frac{1}{2} \partial_i \eta. \quad (1.5)$$

1.2 Prelimeries

Standard definitions of estimation algebras and the Wong matrix may be found in Ocone [1], Shi–Yau [2], and Chapter 2 of Jiao’s doctoral dissertation [4, pp. 21–26].

Shi and Yau proved that the entries of the Wong matrix of a finite-dimensional estimation algebra of state dimension three and linear rank two are polynomials of degree at most one [2, Theorems 3.4 and 3.10]. They also proved the following theorem:

Theorem 1.1. [3, Theorem 1.1, p. 2178]. *Let E be the finite dimensional estimation algebra of with state dimension 3 and rank 2. Then the Ω -matrix has linear structure; i.e., all the Ω -matrix are degree 1 polynomials.*

Thus the genuinely unresolved case is precisely the branch of quadratic rank zero treated here. Jiao investigated the remaining rank-two case from the viewpoint of the structure of the drift and the associated potential function. Jiao further proved that the potential function admits a restricted quartic form. More precisely, Theorem 5.4 of [4] shows that

$$\eta(x) = p_4(x) + \phi(x_3),$$

where

$$p_4 \in \mathbb{R}[x_1, x_2, x_3], \quad \deg p_4 \leq 4.$$

This implies strong algebraic restrictions on the nonlinear drift.

Furthermore, Theorem 5.5 of [4] gives an equivalent description of the drift structure: the drift vector field can be decomposed as

$$f = q + \nabla w,$$

where q is a quadratic polynomial vector field and w is a smooth potential function. Thus, finite-dimensionality together with the rank assumption forces the drift to have a quadratic-polynomial component plus a gradient component.

Jiao then imposed two additional assumptions: the quadratic polynomial drift condition and an elimination condition introduced in Chapter 5. Under these extra hypotheses, Theorem 5.12 of [4] proves that the corresponding filter is a Yau filter, which means that the Wong matrix is constant:

$$\Omega(x) \equiv \Omega_0.$$

Therefore, Jiao's result establishes the constancy of the Wong matrix for a restricted subclass of the rank-two case. The purpose of the present work is to remove these additional assumptions and prove the constancy of the Wong matrix in the general quadratic-rank-zero case.

Theorem 1.2 (Ocone's function-element theorem). *If E is a finite-dimensional estimation algebra and a multiplication operator ϕ belongs to E , then ϕ is a polynomial of degree at most two.*

This is due to Ocone [1]; see also Shi–Yau [2, Theorem 2.7, p. 4230] and Jiao [4, Theorem 2.4, p. 25].

Theorem 1.3 (Polynomiality of highest-order coefficients). *Let E be a finite-dimensional estimation algebra. If, for some $\ell \geq 0$,*

$$A = \sum_{|\alpha|=\ell+1} a_\alpha(x) D^\alpha \pmod{\mathcal{U}_\ell}$$

belongs to E , then every highest-order coefficient a_α is a polynomial.

This is the standard result restated by Shi–Yau [2, Theorem 2.9, p. 4230]. We apply it only to genuine highest-order coefficients and never use it to control the zeroth-order term of a first-order operator.

Theorem 1.4 (Shi–Yau affine structure). *Suppose that E is finite-dimensional, the state dimension is three, and the linear rank is two. After two independent linear functions have been orthonormalized to x_1, x_2 ,*

$$\omega_{12} \in \mathbb{R}[x_1, x_2], \quad \deg \omega_{12} \leq 1,$$

and ω_{13}, ω_{23} are affine polynomials in x_1, x_2, x_3 .

The first assertion is [2, Theorem 3.4, p. 4233]; the latter two are [2, Theorem 3.10, pp. 4243–4245].

Corollary 1.5 (The function-element space). *Under the assumptions of the main theorem, an initial orthogonal change of coordinates can be made so that*

$$E \cap C^\infty(\mathbb{R}^3) = \text{span}_{\mathbb{R}}\{1, x_1, x_2\}. \quad (1.6)$$

Moreover, $x_1, x_2, D_1, D_2, 1 \in E$.

Proof. Since the linear rank is two, choose an orthonormal basis of $\mathcal{L}(E)$ and then make a constant orthogonal change of coordinates so that $x_1, x_2 \in E$. Hence

$$D_i = [L_0, x_i], \quad [D_i, x_i] = 1 \quad (i = 1, 2),$$

and therefore $D_1, D_2, 1 \in E$. On the other hand, by [theorem 1.2](#), every function element is a polynomial of degree at most two, while quadratic rank zero excludes its degree-two part. It can thus be written as $c + \ell$, where $c \in \mathbb{R}$ and ℓ is a homogeneous linear polynomial. Since $1 \in E$, $\ell = (c + \ell) - c \cdot 1 \in \mathcal{L}(E)$, and therefore $\ell \in \text{span}\{x_1, x_2\}$. This proves (1.6) and the membership of all the listed elements.

1.3 Affine parameters, the Bianchi identity, and a gauge representative

By [theorem 1.4](#), initially write

$$\omega_{23} = h_1 x_1 + h_2 x_2 + h_3 x_3 + h_0.$$

Here h_0, h_1, h_2, h_3 are real coefficients of a Wong entry; the observation components have uniformly been denoted by q_j . Because ω is an exterior derivative, one always has

$$\partial_3 \omega_{12} - \partial_2 \omega_{13} + \partial_1 \omega_{23} = 0. \quad (1.7)$$

Note that Since ω_{12} is independent of x_3 , $\partial_3 \omega_{12} = 0$, $\partial_2 \omega_{13} = k_2$, $\partial_3 \omega_{23} = h_1$. It follows that $h_1 = k_2$, which gives (0.3)–(0.5).

On $\mathbb{R}^3$, two smooth drifts with the same curl differ by a gradient. Gauge conjugation

$$\mathcal{G}_\Lambda(P) = e^\Lambda P e^{-\Lambda}$$

satisfies

$$\mathcal{G}_\Lambda(D_i) = D_i - \partial_i \Lambda;$$

it therefore adds $\nabla \Lambda$ to the drift, fixes every multiplication operator, and preserves Lie brackets. We may consequently choose the representative

$$\begin{aligned} f_1 &= 0, \\ f_2 &= \frac{b_1}{2} x_1^2 + b_2 x_1 x_2 + b_0 x_1, \end{aligned} \quad (1.8)$$

$$f_3 = \frac{k_1}{2} x_1^2 + k_2 x_1 x_2 + k_3 x_1 x_3 + k_0 x_1 + \frac{h_2}{2} x_2^2 + h_3 x_2 x_3 + h_0 x_2. \quad (1.9)$$

This is also the short Poincare argument of Jiao's Theorem 5.5 [4, pp. 83–85], reproduced here in full. Gauge conjugation is used only as an isomorphism of Lie algebras of differential operators; we do not claim that the conjugated algebra is the estimation algebra of the same realizable filtering system. $\square$

2 Proof of constancy of ω_{12}

Theorem 2.1. *By direct calculation, we have the following elementary results:*

- We have

$$H_0 = [L_0, D_1] - \frac{1}{2}\partial_2\omega_{12} - \frac{1}{2}\partial_3\omega_{13} = \omega_{12}D_2 + \omega_{13}D_3 + \frac{1}{2}\eta_{11}, \quad (2.1)$$

$$H_1 = [L_0, D_2] - \frac{1}{2}\partial_1\omega_{21} - \frac{1}{2}\partial_3\omega_{23} = \omega_{21}D_1 + \omega_{23}D_3 + \frac{1}{2}\eta_{22}. \quad (2.2)$$

- Set $X_1 = [L_0, H_1]$ and $X_2 = [L_0, H_2]$, we have

$$\begin{aligned} X_0 &= b_1D_1D_2 + b_2D_2^2 + k_1D_1D_3 + k_2D_2D_3 + k_3D_3^2 \\ &\quad + \left(-\omega_{12}^2 - \omega_{13}^2 + \frac{1}{2}\eta_{11}\right)D_1 + \left(-\omega_{13}\omega_{23} + \frac{1}{2}\eta_{12}\right)D_2 \\ &\quad + \left(\omega_{12}\omega_{23} + \frac{1}{2}\eta_{13}\right)D_3 \pmod{\mathcal{U}_0}, \end{aligned} \quad (2.3)$$

$$\begin{aligned} X_1 &= -b_1D_1^2 - b_2D_2D_1 + k_2D_1D_3 + h_2D_2D_3 + h_3D_3^2 \\ &\quad + \left(-\omega_{23}\omega_{13} + \frac{1}{2}\eta_{12}\right)D_1 + \left(-\omega_{21}^2 - \omega_{23}^2 + \frac{1}{2}\eta_{22}\right)D_2 \\ &\quad + \left(-\omega_{21}\omega_{31} + \frac{1}{2}\eta_{23}\right)D_3 \pmod{\mathcal{U}_0}. \end{aligned} \quad (2.4)$$

- Set $Y_0 = [L_0, X_0]$ and $Y_1 = [L_0, X_1]$, then Y_0 and Y_1 are second-order operators:

$$\begin{aligned} Y_0 &= \left(-3k_1\omega_{13} - 3b_1\omega_{12} + \frac{1}{2}\eta_{111}\right)D_1^2 \\ &\quad + \left(b_1\omega_{12} - h_2\omega_{13} - 2k_2\omega_{23} + \frac{1}{2}\eta_{122}\right)D_2^2 \\ &\quad + (-4b_2\omega_{12} - (3k_2 + h_1)\omega_{13} - 2k_1\omega_{23} + \eta_{112})D_1D_2 \\ &\quad + \left(h_3\omega_{12} + k_1\omega_{13} + k_2\omega_{23} + \frac{1}{2}\eta_{133}\right)D_3^2 \\ &\quad + ((h_1 - k_2)\omega_{12} - 4k_3\omega_{13} + 2b_1\omega_{23} + \eta_{113})D_1D_3 \\ &\quad + ((h_2 + k_1)\omega_{12} + (b_1 - h_3)\omega_{13} + 3(b_2 - k_3)\omega_{23} + \eta_{123})D_2D_3 \pmod{\mathcal{U}_1}. \end{aligned} \quad (2.5)$$

$$\begin{aligned} Y_1 &= \left(b_2\omega_{12} - 2k_2\omega_{13} - k_1\omega_{23} + \frac{1}{2}\eta_{112}\right)D_1^2 \\ &\quad + \left(-3b_2\omega_{12} - 3h_2\omega_{23} + \frac{1}{2}\eta_{222}\right)D_2^2 \\ &\quad + (-4b_1\omega_{12} - 2h_2\omega_{13} - 4k_2\omega_{23} + \eta_{122})D_1D_2 \\ &\quad + \left(-k_3\omega_{12} + k_2\omega_{13} + h_2\omega_{23} + \frac{1}{2}\eta_{233}\right)D_3^2 \\ &\quad + (-2b_2\omega_{13} - 4h_3\omega_{23} + \eta_{223})D_2D_3 \\ &\quad + (-(h_2 + k_1)\omega_{12} - 3(b_1 + h_3)\omega_{13} - (b_2 + k_3)\omega_{23} + \eta_{123})D_1D_3 \pmod{\mathcal{U}_1}. \end{aligned} \quad (2.6)$$

2.1 The visible second-order class and triangular growth

Definition 2.2. Let $\mathcal{G}$ be the set of all elements of E of order at most two. For $B \in \mathcal{G}$, write its second-order part as

$$B = a_{11}D_1^2 + a_{22}D_2^2 + a_{33}D_3^2 + a_{12}D_1D_2 + a_{13}D_1D_3 + a_{23}D_2D_3 \pmod{\mathcal{U}_1}. \quad (2.7)$$

Note that since $B \in E$,

$$[[B, x_1], x_1] = 2a_{11} \in E, \quad [[B, x_2], x_2] = 2a_{22} \in E, \quad [[B, x_1], x_2] = a_{12} \in E.$$

It follows from (1.6) that

$$a_{11}, a_{22}, a_{12} \in \text{span}_{\mathbb{R}}\{1, x_1, x_2\}.$$

Theorem 2.3. Let $B \in \mathcal{G}$, and write $B = Q_B + R_B \pmod{\mathcal{U}_1}$:

$$Q_B = a_{11}D_1^2 + a_{12}D_1D_2 + a_{22}D_2^2 \quad (2.8)$$

$$R_B = a_{33}D_3^2 + a_{13}D_1D_3 + a_{23}D_2D_3, \quad (2.9)$$

Then for

$$[H_0, B] = [\omega_{12}D_2, Q_B] + [\omega_{12}D_2, R_B] + [\omega_{13}D_3, Q_B] + [\omega_{13}D_3, R_B] \pmod{\mathcal{U}_1}. \quad (2.10)$$

we have the following elementary results:

- (1). $[H_0, B] \in \mathcal{G}$;
- (2). All $[\omega_{12}D_2, R_B]$, $[\omega_{13}D_3, Q_B]$, and $[\omega_{13}D_3, R_B]$ contain operator D_3 ;
- (3). We have

$$[\omega_{12}D_2, Q_B] = (\omega_{12}\partial_2a_{22} - b_1a_{12} - 2b_2a_{22})D_2^2 \quad (2.11)$$

$$+ (\omega_{12}\partial_2a_{11})D_1^2 + (\omega_{12}\partial_2a_{12} - 2b_1a_{11} - b_2a_{12})D_1D_2. \pmod{\mathcal{U}_1} \quad (2.12)$$

Similarly, $[H_1, B] \in \mathcal{G}$, all $[\omega_{21}D_1, R_B]$, $[\omega_{23}D_3, Q_B]$, and $[\omega_{13}D_3, Q_B]$ contain operator D_3 , while

$$[\omega_{21}D_1, Q_B] = (-\omega_{12}\partial_1a_{11} + 2b_1a_{11} + b_2a_{12})D_1^2 \quad (2.13)$$

$$- \omega_{12}\partial_1a_{22}D_2^2 + (-\omega_{12}\partial_1a_{12} + b_1a_{12} + 2b_2a_{22})D_1D_2 \pmod{\mathcal{U}_1}. \quad (2.14)$$

Proof of Theorem 2.3

Proof. We only prove the assertion for H_0 . Put $w = \omega_{12}, u = \omega_{13}$. Thus

$$\omega_{21} = -w, \quad \partial_1w = b_1, \quad \partial_2w = b_2, \quad \partial_3w = 0.$$

Moreover, since $B \in \mathcal{G}$, Definition 2.2 gives

$$a_{11}, a_{12}, a_{22} \in \text{Span}_{\mathbb{R}}\{1, x_1, x_2\},$$

and hence

$$\partial_3a_{11} = \partial_3a_{12} = \partial_3a_{22} = 0.$$

Proof of the first assertion. The order filtration satisfies

$$[\mathcal{U}_r, \mathcal{U}_s] \subseteq \mathcal{U}_{r+s-1}.$$

By construction, $H_0, H_1 \in E \cap \mathcal{U}_1$, while $B \in E \cap \mathcal{U}_2$. Since E is a Lie algebra,

$$[H_i, B] \in E \cap \mathcal{U}_2 = \mathcal{G}, \quad i = 0, 1.$$

Furthermore, for some $\phi_0, \phi_1 \in \mathcal{U}_0$,

$$H_0 = wD_2 + uD_3 + \phi_0, \quad H_1 = -wD_1 + vD_3 + \phi_1.$$

Since $B = Q_B + R_B \pmod{\mathcal{U}_1}$, the filtration identities

$$[\mathcal{U}_0, \mathcal{U}_2] \subseteq \mathcal{U}_1, \quad [\mathcal{U}_1, \mathcal{U}_1] \subseteq \mathcal{U}_1$$

give

$$\begin{aligned} [H_0, B] &= [wD_2, Q_B] + [wD_2, R_B] \\ &\quad + [uD_3, Q_B] + [uD_3, R_B] \pmod{\mathcal{U}_1}, \end{aligned}$$

and similarly

$$\begin{aligned} [H_1, B] &= [-wD_1, Q_B] + [-wD_1, R_B] \\ &\quad + [vD_3, Q_B] + [vD_3, R_B] \pmod{\mathcal{U}_1}. \end{aligned}$$

For the remaining assertions, we use the elementary identity

$$[fD_\ell, aD_iD_j] = f(\partial_\ell a)D_iD_j - a(\partial_i f)D_\ell D_j - a(\partial_j f)D_iD_\ell \pmod{\mathcal{U}_1}. \quad (*)$$

Proof of the second assertion. Define

$$\mathcal{I}_3^{(2)} := \{c_{13}D_1D_3 + c_{23}D_2D_3 + c_{33}D_3^2 : c_{ij} \in C^\infty(\mathbb{R}^3)\} + \mathcal{U}_1.$$

Thus $P \in \mathcal{I}_3^{(2)}$ precisely means that every monomial in the second-order part of P contains D_3 . Applying $(*)$ to $[wD_2, R_B]$, the only terms in which the D_3 -factor could disappear are multiplied by $\partial_3 w = 0$. Hence

$$[wD_2, R_B] \in \mathcal{I}_3^{(2)}.$$

For $[uD_3, Q_B]$, the transport term in $(*)$ vanishes because the coefficients of Q_B are independent of x_3 , while each remaining term contains D_3 . Therefore

$$[uD_3, Q_B] \in \mathcal{I}_3^{(2)}.$$

Finally, in $[uD_3, R_B]$, the transport term retains the original D_3 -factor and each of the other terms introduces a D_3 -factor. Consequently,

$$[uD_3, R_B] \in \mathcal{I}_3^{(2)}.$$

Proof of the third assertion. Applying $(*)$ term by term gives

$$\begin{aligned} [wD_2, a_{11}D_1^2] &= (w\partial_2 a_{11})D_1^2 - 2b_1a_{11}D_1D_2, \\ [wD_2, a_{12}D_1D_2] &= -b_1a_{12}D_2^2 + (w\partial_2 a_{12} - b_2a_{12})D_1D_2, \\ [wD_2, a_{22}D_2^2] &= (w\partial_2 a_{22} - 2b_2a_{22})D_2^2 \pmod{\mathcal{U}_1}. \end{aligned}$$

Adding these identities yields

$$\begin{aligned} [wD_2, Q_B] &\equiv (w\partial_2 a_{11})D_1^2 \\ &\quad + (w\partial_2 a_{22} - b_1a_{12} - 2b_2a_{22})D_2^2 \\ &\quad + (w\partial_2 a_{12} - 2b_1a_{11} - b_2a_{12})D_1D_2 \pmod{\mathcal{U}_1}. \end{aligned}$$

□

2.2 Classification of the $\mathcal{G}$ coefficients and final elimination of the slopes of ω_{12}

Lemma 2.4. *Let $C \in \mathcal{G}$.*

- (i) *If $[D_1^2][C] = Ax_1 + C_0$, then $A = 0$.*
- (ii) *If $[D_2^2][C] = Bx_2 + C_0$, then $B = 0$.*

Proof of the pure-head Lie-growth lemma

Proof. Define

$$\mathcal{T}_m^{(1)} := \left\{ \sum_{\substack{|\alpha|=m \\ \alpha_2+\alpha_3 \geq 1}} r_\alpha(x) D^\alpha \right\}. \quad (2)$$

Thus $\mathcal{U}_{m-1} + \mathcal{T}_m^{(1)}$ contains every term of order at most $m - 1$ and every order- m monomial containing D_2 or D_3 .

Put $u = Ax_1 + C_0$. Then

$$\partial_1 u = A, \quad \partial_2 u = \partial_3 u = 0. \quad (3)$$

The second-order part of C has the decomposition

$$C = uD_1^2 + S \pmod{\mathcal{U}_1}, \quad S \in \mathcal{T}_2^{(1)}. \quad (4)$$

Equations (1)–(4) give, for every $m \geq 1$,

$$[\mathcal{T}_m^{(1)}, uD_1^2] \subseteq \mathcal{T}_{m+1}^{(1)} + \mathcal{U}_m, \quad (5)$$

$$[\mathcal{T}_m^{(1)}, \mathcal{T}_2^{(1)}] \subseteq \mathcal{T}_{m+1}^{(1)} + \mathcal{U}_m, \quad (6)$$

and, for every $c \in \mathbb{R}$,

$$[cD_1^m, \mathcal{T}_2^{(1)}] \subseteq \mathcal{T}_{m+1}^{(1)} + \mathcal{U}_m. \quad (7)$$

Define the iteration constructure

$$P_0 = [L_0, C], \quad P_{r+1} = [P_r, C], \quad (9)$$

equations (3)–(7) give

$$\begin{aligned} P_0 &= \left[\frac{1}{2} D_1^2, uD_1^2 \right] + R_0 \pmod{\mathcal{U}_2} \\ &= AD_1^3 + R_0 \pmod{\mathcal{U}_2}, \quad R_0 \in \mathcal{T}_3^{(1)}. \end{aligned} \quad (11)$$

Assume

$$P_r = c_r D_1^{r+3} + R_r \pmod{\mathcal{U}_{r+2}}, \quad R_r \in \mathcal{T}_{r+3}^{(1)}. \quad (12)$$

Then

$$\begin{aligned} P_{r+1} &= [c_r D_1^{r+3}, uD_1^2] + [c_r D_1^{r+3}, S] \\ &\quad + [R_r, uD_1^2] + [R_r, S] \pmod{\mathcal{U}_{r+3}}. \end{aligned} \quad (13)$$

Equations (5)–(8) imply

$$P_{r+1} = (r+3)Ac_r D_1^{r+4} + R_{r+1} \pmod{\mathcal{U}_{r+3}}, \quad (14)$$

where

$$R_{r+1} \in \mathcal{T}_{r+4}^{(1)}. \quad (15)$$

Consequently,

$$c_0 = A, \quad c_{r+1} = (r+3)Ac_r, \quad (16)$$

and hence

$$c_r = \frac{(r+2)!}{2} A^{r+1}. \quad (17)$$

If $A \neq 0$, then the the following iterations P_{r+1} have terms D_1^{r+3} with nonzero coefficients, which is a contradiction to the finite-dimensionality of E . Therefore $A = 0$.

Similarly for the second assertion $B = 0$. $\square$

Theorem 2.5. *If $(b_1, b_2) \neq (0, 0)$, then for every $B \in \mathcal{G}$ there are real constants λ, μ, γ such that*

$$a_{11} = \lambda\omega_{12} + c_1, \quad a_{22} = \mu\omega_{12} + c_2, \quad a_{12} = \gamma\omega_{12} + c_{12}. \quad (2.15)$$

Proof of Theorem 2.5

Proof. Put

$$w := \omega_{12} = b_1x_1 + b_2x_2 + b_0, \quad (b_1, b_2) \neq (0, 0). \quad (30)$$

Step 1: Write $a_{11} = \alpha_1x_1 + \alpha_2x_2 + \alpha_0$. For $B \in \mathcal{G}$, we define $B_{11} := [H_0, B] - b_2B \in \mathcal{G}$. Using (35),

$$\begin{aligned} [D_1^2][B_{11}] &= w\partial_2a_{11} - b_2a_{11} \\ &= \alpha_2w - b_2(\alpha_1x_1 + \alpha_2x_2 + \alpha_0) \\ &= (\alpha_2b_1 - \alpha_1b_2)x_1 + (\alpha_2b_0 - b_2\alpha_0). \end{aligned}$$

Lemma 2.4(i) gives

$$\alpha_2b_1 - \alpha_1b_2 = 0.$$

Therefore $(\alpha_1, \alpha_2) = \lambda(b_1, b_2)$ for some $\lambda \in \mathbb{R}$, and one obtains

$$\begin{aligned} a_{11} &= \lambda b_1x_1 + \lambda b_2x_2 + \alpha_0 \\ &= \lambda w + (\alpha_0 - \lambda b_0). \end{aligned}$$

Step 2: Write $a_{22} = \beta_1x_1 + \beta_2x_2 + \beta_0$. Define $B_{22} := [H_1, B] + b_1B \in \mathcal{G}$. Using (36),

$$\begin{aligned} [D_2^2][B_{22}] &= -w\partial_1a_{22} + b_1a_{22} \\ &= -\beta_1w + b_1(\beta_1x_1 + \beta_2x_2 + \beta_0) \\ &= (b_1\beta_2 - b_2\beta_1)x_2 + (b_1\beta_0 - b_0\beta_1). \end{aligned}$$

Lemma 2.4(ii) gives

$$b_1\beta_2 - b_2\beta_1 = 0.$$

Therefore $(\beta_1, \beta_2) = \mu(b_1, b_2)$ for some $\mu \in \mathbb{R}$, and one obtains

$$\begin{aligned} a_{22} &= \mu b_1x_1 + \mu b_2x_2 + \beta_0 \\ &= \mu w + (\beta_0 - \mu b_0). \end{aligned}$$

Step 3: From (47) and (56),

$$\partial_1a_{11} = \lambda b_1, \quad \partial_2a_{22} = \mu b_2. \quad (57)$$

If $b_2 \neq 0$, put

$$C_1 := [H_1, B] \in \mathcal{G}. \quad (59)$$

Equation (47), applied to C_1 , gives

$$[D_1^2][C_1] = \Lambda w + C \quad (60)$$

for some $\Lambda, C \in \mathbb{R}$. Equations (37), (47), and (57) give

$$\begin{aligned} \Lambda w + C &= -w(\lambda b_1) + 2b_1(\lambda w + c_1) + b_2a_{12} \\ &= \lambda b_1w + 2b_1c_1 + b_2a_{12}. \end{aligned} \quad (61)$$

Hence

$$b_2a_{12} = (\Lambda - \lambda b_1)w + (C - 2b_1c_1), \quad (62)$$

and therefore

$$a_{12} = \frac{\Lambda - \lambda b_1}{b_2} w + \frac{C - 2b_1c_1}{b_2}. \quad (63)$$

If $b_1 \neq 0$ we do the same procedure and yield the same result. $\square$

Lemma 2.6. *If ω_{12} is nonconstant, then $b_1 b_2 \neq 0$.*

Exclusion of the one-axis branches

Proof. Assume $b_1 = 0, b_2 \neq 0$, and take $B = Y_0$. By Theorem 2.5 and lemma $\mu b_2 = 0$, hence $\mu = 0$. B can be formulated as:

$$B = (\lambda b_2 x_2 + c_1) D_1^2 + c_2 D_2^2 + (\gamma b_2 x_2 + c_{12}) D_1 D_2 + R_B \pmod{\mathcal{U}_1}, \quad (2.16)$$

Set $P_0 = [L_0, B]$, $P_{r+1} = [P_r, B]$, and define $c_r := [D_1^{r+1} D_2^2][P_r]$. For $r = 0$,

$$\begin{aligned} c_0 &= \partial_2 a_{12} + \partial_1 a_{22} \\ &= \partial_2 (\gamma b_2 x_2 + c_{12}) + \partial_1 c_2 \\ &= \gamma b_2. \end{aligned}$$

Suppose

$$[D_1^{r+1} D_2^2][P_r] = c_r.$$

Since $[D_1^{r+1} D_2^2, x_2] = 2D_1^{r+1} D_2$, one has

$$[c_r D_1^{r+1} D_2^2, \gamma b_2 x_2 D_1 D_2] = 2\gamma b_2 c_r D_1^{r+2} D_2^2 \pmod{\mathcal{U}_{r+3}}.$$

Hence

$$c_{r+1} = 2\gamma b_2 c_r.$$

The λ -term decreases the D_2 -exponent, the constant terms produce no new highest-order term, and terms involving R_B retain a factor D_3 . Thus no other term contributes to the tracked slot. Therefore

$$c_r = (2\gamma b_2)^r c_0 = 2^r (\gamma b_2)^{r+1},$$

If $\gamma \neq 0$, then $\text{ord}(P_r) = r + 3 \rightarrow \infty$, contradicting finite-dimensionality. Therefore $\gamma = 0$. Thus the D_2^2 - and $D_1 D_2$ -coefficients of Y_0 are constant.

From

$$[D_2^2][Y_0] = -h_2 \omega_{13} - 2k_2 \omega_{23} + \frac{1}{2} \eta_{122},$$

differentiation by x_1 gives

$$0 = -h_2 k_1 - 2k_2 h_1 + \frac{1}{2} \eta_{1122}.$$

Using $h_1 = k_2$,

$$\frac{1}{2} \eta_{1122} = h_2 k_1 + 2k_2^2.$$

From

$$[D_1 D_2][Y_0] = -4b_2 \omega_{12} - (3k_2 + h_1) \omega_{13} - 2k_1 \omega_{23} + \eta_{112},$$

differentiation by x_2 gives

$$0 = -4b_2^2 - (3k_2 + h_1) k_2 - 2k_1 h_2 + \eta_{1122}.$$

Again using $h_1 = k_2$,

$$\eta_{1122} = 4b_2^2 + 4k_2^2 + 2k_1 h_2.$$

Doubling the first identity and comparing yields $4b_2^2 = 0$, contrary to $b_2 \neq 0$.

The branch $b_2 = 0, b_1 \neq 0$ follows similarly. □

Theorem 2.7. *Under the main assumptions,*

$$b_1 = b_2 = 0, \quad \omega_{12} = b_0.$$

Proof of Theorem 2.7

Proof. Suppose, to the contrary, that ω_{12} is nonconstant. By lemma 2.6, $b_1 b_2 \neq 0$. Let $B \in \mathcal{G}$ be arbitrary. By theorem 2.5, write $\omega_{12} = w$:

$$a_{11} = \lambda\omega + c_1, \quad a_{22} = \mu\omega + c_2, \quad a_{12} = \gamma\omega + c_{12}.$$

For $B' = [H_0, B] - b_2 B$, $[D_1^2][B'] = -b_2 c_1$ is a constant, hence $\lambda' = 0$.

$$\partial_w \lambda = 0, \quad \partial_w \mu = -2\mu b_2 - \gamma b_1, \quad \partial_w \gamma = -2\lambda b_1 - \gamma b_2.$$

By lemma,

$$\partial_w \mu = 0.$$

If $\partial_w \gamma \neq 0$, then like above we can construct a sequence of iterations P_r such that $[D_1^{r+2} D_2][P_r] = b_1 (\gamma')^{r+1} b_2^r$, which is a contradiction to the finite-dimensionality. Therefore,

$$2\mu b_2 + \gamma b_1 = 0, \quad 2\lambda b_1 + \gamma b_2 = 0.$$

Since $b_1 b_2 \neq 0$, define $\kappa := \frac{\gamma}{2b_1 b_2}$. Then

$$\gamma = 2b_1 b_2 \kappa, \mu = -b_1^2 \kappa, \lambda = -b_2^2 \kappa.$$

Now take $B = Y_0$. The three visible coefficients in (2.5) give

$$-3k_1 \omega_{13} - 3b_1 \omega_{12} + \frac{1}{2} \eta_{111} = \lambda \omega_{12} + c_1, \quad (2.17)$$

$$b_1 \omega_{12} - h_2 \omega_{13} - 2k_2 \omega_{23} + \frac{1}{2} \eta_{122} = \mu \omega_{12} + c_2, \quad (2.18)$$

$$-4b_2 \omega_{12} - (3k_2 + h_1) \omega_{13} - 2k_1 \omega_{23} + \eta_{112} = \gamma \omega_{12} + c_{12}. \quad (2.19)$$

Differentiate (2.17) by ∂_2 , (2.18) by ∂_1 , and (2.19) by ∂_1 and ∂_2 , respectively. This gives

$$-3k_1 k_2 - 3b_1 b_2 + \frac{1}{2} \eta_{1112} = -b_2^3 \kappa, \quad (2.20)$$

$$b_1^2 - h_2 k_1 - 2k_2 h_1 + \frac{1}{2} \eta_{1122} = -b_1^3 \kappa, \quad (2.21)$$

$$-4b_1 b_2 - (3k_2 + h_1) k_1 - 2k_1 h_1 + \eta_{1112} = 2b_1^2 b_2 \kappa, \quad (2.22)$$

$$-4b_2^2 - (3k_2 + h_1) k_2 - 2k_1 h_2 + \eta_{1122} = 2b_1 b_2^2 \kappa. \quad (2.23)$$

Using $h_1 = k_2$ and comparing the two pairs containing the same fourth-order derivatives gives

$$\kappa = \frac{b_1}{b_1^2 + b_2^2}, \quad \kappa = -\frac{b_1^2 + 2b_2^2}{b_1(b_1^2 + b_2^2)}.$$

Consequently, $b_1^2 + b_2^2 = 0$, contradicting $b_1 b_2 \neq 0$. Therefore we prove $b_1 = b_2 = 0, \omega_{12} = b_0$. $\square$

3 Proof of $k_3 = h_3 = 0$

3.1 Necessary Lemmas

Lemma 3.1 (Global smooth Euler equation). *Let $Q \in \mathbb{C}[s] \setminus \{0\}$ and let $A = t\partial_t$. Then*

$$\ker_{C^\infty(\mathbb{R}; \mathbb{C})} Q(A) = \text{span}_{\mathbb{C}} \{t^n : n \in \mathbb{N}_0, Q(n) = 0\}.$$

In particular, every globally smooth solution of an Euler equation is a finite polynomial.

Proof of Lemma 3.1

Proof. For $u \in C^\infty(\mathbb{R}; \mathbb{C})$, the Leibniz rule gives

$$(Q(A)u)^{(n)}(0) = Q(n)u^{(n)}(0).$$

Suppose that $Q(A)u = 0$, and set

$$S = \{n \in \mathbb{N}_0 : Q(n) = 0\}, \quad p(t) = \sum_{n \in S} \frac{u^{(n)}(0)}{n!} t^n.$$

The set S is finite, $Q(A)p = 0$, and $v = u - p$ satisfies $Q(A)v = 0$ and is flat at $t = 0$.

We now rigorously exclude nonzero flat solutions. On $t > 0$, set $t = e^s$ and $V_+(s) = v(e^s)$; on $t < 0$, set $t = -e^s$ and $V_-(s) = v(-e^s)$. On both half-lines, $A = \partial_s$, and hence

$$Q(\partial_s)V_\pm = 0.$$

By ODE theory, if the distinct complex roots of Q are μ_j , with multiplicities m_j , the solutions of this constant-coefficient equation are

$$V_\pm(s) = \sum_j e^{\mu_j s} P_j^\pm(s), \quad \deg P_j^\pm < m_j.$$

The function v is flat at $t = 0$, and $v(t) = o(|t|^L)$ for any $L > 0$,

$$\partial_s^k V_\pm(s) = o(e^{Ls}) \quad (s \rightarrow -\infty).$$

If some $P_j^\pm \neq 0$, then we can choose $L > \max_{j: P_j^\pm \neq 0} \text{Re } \mu_j$ contradicts the estimate $O(e^{Ls})$. Hence $V_+ = V_- = 0$, and consequently $v = 0$. $\square$

Lemma 3.2 (Finite-dimensional Euler decomposition). *Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra of finite-order differential operators on $\mathbb{R}_y^2 \times \mathbb{R}_t$, with globally smooth coefficients. Suppose that*

$$J_0 = \lambda t \partial_t + r_0(y) \in \mathfrak{g}, \quad \lambda \in \mathbb{R} \setminus \{0\}, \quad r_0 \in C^\infty(\mathbb{R}^2).$$

Every $P \in \mathfrak{g}$ has a unique normal form

$$P = \sum_{\alpha \in \mathbb{N}_0^2, q \in \mathbb{N}_0} a_{\alpha q}(y, t) \partial_{y_1}^{\alpha_1} \partial_{y_2}^{\alpha_2} \partial_t^q.$$

Then

- every coefficient satisfies

$$a_{\alpha q}(y, t) \in C^\infty(\mathbb{R}^2)[t].$$

- Write

$$a_{\alpha q}(y, t) = \sum_{p \geq 0} a_{\alpha q p}(y) t^p$$

and define

$$\text{wt} \left(a_{\alpha q p}(y) t^p \partial_{y_1}^{\alpha_1} \partial_{y_2}^{\alpha_2} \partial_t^q \right) := p - q.$$

Then, only finitely many weights occur. If P_w denotes the sum of all terms of P having weight w , then there exists $\pi_w \in \mathbb{R}[s]$ such that

$$P_w = \pi_w(\text{ad}_{J_0})P \in \mathfrak{g}, \quad \text{ad}_{J_0}(P) := [J_0, P].$$

Thus every weight component is obtained from actual iterated Lie brackets with J_0 .

Proof of Lemma 3.2

Define

$$T(P) := [J_0, P], \quad D(P) := [\lambda t \partial_t, P], \quad N(P) := [r_0(y), P].$$

Thus

$$T = D + N.$$

Since r_0 is independent of t ,

$$[\lambda t \partial_t, r_0] = 0.$$

For a normal-form term $a(y, t) \partial_y^\alpha \partial_t^q$, $\partial_y^\alpha = \partial_{y_1}^{\alpha_1} \partial_{y_2}^{\alpha_2}$, direct calculation gives

$$D \left(a \partial_y^\alpha \partial_t^q \right) = \lambda (t \partial_t - q) a \partial_y^\alpha \partial_t^q$$

Hence D preserves the y -differential order, whereas every application of N lowers it by at least one.

Choose a finite basis of $\mathfrak{g}$. Let K be the largest y -differential order appearing in that basis, then

$$N^{K+1}P = 0 \quad (P \in \mathfrak{g}).$$

Because $J_0 \in \mathfrak{g}$ and $\mathfrak{g}$ is a Lie algebra, $T = \text{ad}_{J_0}$ is a linear endomorphism of a finite-dimensional vector space. Let χ be its characteristic polynomial. By the Cayley–Hamilton theorem,

$$\chi(T)P = 0 \quad (P \in \mathfrak{g}).$$

Since D and N commute by Jacobi identity, there is a polynomial R in two commuting variables such that

$$\chi(D + N) = \chi(D) + NR(D, N).$$

All operators in this identity commute. Hence

$$\chi(D)^{K+1}P = (\chi(T) - NR(D, N))^{K+1}P = 0.$$

Indeed, every binomial term containing $\chi(T)$ can be rearranged so that $\chi(T)$ acts directly on P and therefore vanishes. The remaining term contains N^{K+1} and also vanishes.

Now write

$$P = \sum_{\alpha, q} a_{\alpha q}(y, t) \partial_y^\alpha \partial_t^q.$$

The operator D does not mix different pairs (α, q) . By uniqueness of normal form,

$$[\chi(\lambda(t \partial_t - q))]^{K+1} a_{\alpha q}(y, t) = 0.$$

For fixed y , this is an Euler equation in t . The global smooth Euler lemma therefore gives

$$a_{\alpha q}(y, t) = \sum_{\substack{p \in \mathbb{N}_0 \\ \chi(\lambda(p-q))=0}} \frac{\partial_t^p a_{\alpha q}(y, 0)}{p!} t^p \in C^\infty(\mathbb{R}^2)[t].$$

Thus P has finite sum

$$P = \sum_{w \in \mathbb{Z}, \chi(\lambda w)=0} \sum_{p-q=w} P_w,$$

Note that

$$D(P_w) = \lambda w P_w.$$

Moreover, N preserves the weight. Consequently,

$$(T - \lambda w)^{K+1} P_w = N^{K+1} P_w = 0.$$

The polynomials

$$(s - \lambda w)^{K+1}, \quad w \in \{\chi(\lambda w) = 0\},$$

are pairwise coprime. By the Chinese remainder theorem, for every $w \in W$ there exists $\pi_w \in \mathbb{R}[s]$ such that

$$\pi_w \equiv 1 \pmod{(s - \lambda w)^{K+1}},$$

and

$$\pi_w \equiv 0 \pmod{(s - \lambda v)^{K+1}} \quad (v \in \{w : \chi(\lambda w) = 0\}, v \neq w).$$

It follows that

$$\pi_w(T)P = P_w.$$

Since $T(P) = [J_0, P]$ preserves $\mathfrak{g}$,

$$P_w = \pi_w(\text{ad}_{J_0})P \in \mathfrak{g}.$$

3.2 Proof of $k_3 = h_3 = 0$.

Theorem 3.3. *Under the main theorem's assumptions, we have $k_3 = h_3 = 0$.*

Proof of Theorem 3.3

Suppose $(k_3, h_3) \neq 0$. Define the Poincaré representative f^P by

$$f_1^P = 0, \tag{3.1}$$

$$f_2^P = b_0 x_1, \tag{3.2}$$

$$f_3^P = \frac{k_1}{2} x_1^2 + k_2 x_1 x_2 + k_3 x_1 x_3 + k_0 x_1 + \frac{h_2}{2} x_2^2 + h_3 x_2 x_3 + h_0 x_2. \tag{3.3}$$

Direct differentiation gives that f^P and f have the same Wong matrix. The Poincaré lemma gives $\Phi \in C^\infty(\mathbb{R}^3)$ such that

$$f^P - f = \nabla \Phi.$$

Define

$$\mathcal{G}_\Phi(P) := e^\Phi P e^{-\Phi}, \quad E^P := \mathcal{G}_\Phi(E),$$

we have

$$D_i^P := \mathcal{G}_\Phi(D_i) = \partial_i - f_i^P.$$

Furthermore,

$$L_0^P := \mathcal{G}_\Phi(L_0) = \frac{1}{2} \sum_{i=1}^3 (D_i^P)^2 - \frac{1}{2} \eta.$$

The map $\mathcal{G}_\Phi$ is a Lie-algebra isomorphism. In particular, $D_1^P, D_2^P, L_0^P \in E^P$, and the correspondingly transformed H_0^P, H_1^P belong to E^P .

From now on we drop the superscript P and again write E, D_i, L_0, H_i, η .

Set

$$\alpha = \frac{k_3 k_1 + h_3 k_2}{\mu}, \quad \beta = \frac{k_3 k_2 + h_3 h_2}{\mu}, \quad \delta = \frac{k_3 k_0 + h_3 h_0}{\mu},$$

and

$$t := \frac{k_3 \omega_{13} + h_3 \omega_{23}}{\mu} = x_3 + \alpha x_1 + \beta x_2 + \delta.$$

Let

$$\nabla_1 := \partial_1 - \alpha \partial_3, \quad \nabla_2 := \partial_2 - \beta \partial_3.$$

Then

$$\nabla_1 t = \nabla_2 t = 0, \quad [\nabla_1, \nabla_2] = [\nabla_i, \partial_3] = 0.$$

All normal forms and weights below are taken with respect to this commuting frame.

The formulas for H_0, H_1 give the genuine element

$$\begin{aligned} J &:= \frac{k_3(H_0 - b_0 D_2) + h_3(H_1 + b_0 D_1)}{\mu} \\ &= t D_3 + \frac{k_3 \eta_1 + h_3 \eta_2}{2\mu} + \frac{1}{2} = t \partial_3 + r(x_1, x_2, t) \in E \end{aligned}$$

for some smooth function r . Since $r - r_0$ vanishes on $t = 0$, Hadamard's lemma gives a smooth function Λ satisfying $\partial_3 \Lambda = \frac{r - r_0}{t}$. For

$$\widehat{E} := e^\Lambda E e^{-\Lambda},$$

we therefore have

$$\widehat{J} = e^\Lambda J e^{-\Lambda} = t \partial_3 + r_0(x_1, x_2).$$

We now apply the smooth Euler finite-module lemma. Since

$$\widehat{D}_1 = \nabla_1 + \alpha \partial_3 - \partial_1 \Lambda, \quad \widehat{D}_2 = \nabla_2 + \beta \partial_3 - \partial_2 \Lambda,$$

the Euler finite-module lemma shows that $\partial_1 \Lambda$ and $\partial_2 \Lambda$ are in $C^\infty[x_1, x_2](t)$. Every positive-weight component of either function is a multiplication element of $\widehat{E}$. But

$$\widehat{E} \cap C^\infty(\mathbb{R}^3) = \text{span}_{\mathbb{R}}\{1, x_1, x_2\}$$

has only weight zero. Hence $\partial_1 \Lambda$ and $\partial_2 \Lambda$ are independent of t . Consequently,

$$\partial_i(\partial_3 \Lambda) = \partial_3(\partial_i \Lambda) = 0, \quad i = 1, 2,$$

and therefore

$$\partial_3 \Lambda = \rho(x_3)$$

for some $\rho \in C^\infty(\mathbb{R})$.

The Poincaré formula for f_3 , after substituting $x_3 = t - \alpha x_1 - \beta x_2 - \delta$, has the form

$$f_3 = (k_3 x_1 + h_3 x_2)t + F_0(x_1, x_2).$$

Direct normal ordering of $\widehat{L}_0$ now gives

$$\begin{aligned}\widehat{L}_0 = & \frac{1}{2} \left((\nabla_1 + \alpha \partial_3)^2 + (\nabla_2 + \beta \partial_3)^2 + \partial_3^2 \right) \\ & + a_1(x_1, x_2) \nabla_1 + a_2(x_1, x_2) \nabla_2 \\ & - \left[(k_3 x_1 + h_3 x_2) t + \rho(t - \alpha x_1 - \beta x_2 - \delta) + b(x_1, x_2) \right] \partial_3 \\ & + u(x_1, x_2, t),\end{aligned}\tag{3.4}$$

for some smooth a_1, a_2, b, u .

A second application of the Euler finite-module lemma shows that every coefficient in (3.4) is polynomial in t . In particular, setting $x_1 = x_2 = 0$ in the coefficient of ∂_3 gives

$$\rho(t - \delta) + b(0, 0) \in \mathbb{R}[t].$$

Thus ρ is a one-variable polynomial.

The second-order part of $\widehat{L}_0$ is

$$\frac{1}{2} \nabla_1^2 + \frac{1}{2} \nabla_2^2 + \alpha \nabla_1 \partial_3 + \beta \nabla_2 \partial_3 + \frac{1 + \alpha^2 + \beta^2}{2} \partial_3^2.$$

Its weights are 0, 0, -1 , -1 , -2 , respectively. Every first-order term in (3.4) has weight no smaller than -1 , and every term of u has nonnegative weight. Hence by the Euler projection lemma

$$\partial_3^2 \in \widehat{E}.$$

It remains to obtain unbounded differential order.

(i) If $\deg \rho \geq 2$,

$$\rho(s) = a_d s^d + \cdots, \quad d \geq 2, \quad a_d \neq 0.$$

Then $T := p_{d-1}(\text{ad}_{\widehat{J}})(\widehat{L}_0) = a_d t^d \partial_3 + q(x_1, x_2, t) \in \widehat{E}$.

Assume inductively that $P_m \in \widehat{E}$ satisfies

$$P_m = \partial_3^m \pmod{\mathcal{U}_{m-1}}.$$

Set

$$R_m := [P_m, [T, P_m]] = m^2 a_d d(d-1) t^{d-2} \partial_3^{2m-1} \pmod{\mathcal{U}_{2m-2}},$$

we obtain

$$\text{ad}_{\partial_3^2}^{d-2}(R_m) = 2^{d-2} (d-2)! m^2 a_d d(d-1) \partial_3^{2m+d-3} \pmod{\mathcal{U}_{2m+d-4}}.$$

The displayed coefficient is nonzero, which contradicts to the finite-dimensionality.

(ii) If $\deg \rho = 1$, the weight-zero component of $\widehat{L}_0$ is

$$\begin{aligned}L^{[0]} := \Pi_0(\widehat{L}_0) = & \frac{1}{2} (\nabla_1^2 + \nabla_2^2) + a_1 \nabla_1 + a_2 \nabla_2 \\ & - (k_3 x_1 + h_3 x_2 + c) t \partial_3 + u_0(x_1, x_2).\end{aligned}$$

Assume $\partial_3^m \in \widehat{E}$. Then

$$X_m := \frac{[L^{[0]}, \partial_3^m] - m c \partial_3^m}{m} = (k_3 x_1 + h_3 x_2) \partial_3^m \in \widehat{E}.$$

A direct calculation gives

$$[L^{[0]}, X_m] = (k_3 \nabla_1 + h_3 \nabla_2 + M_m(x_1, x_2)) \partial_3^m,$$

where

$$M_m = k_3 a_1 + h_3 a_2 + m(k_3 x_1 + h_3 x_2 + c)(k_3 x_1 + h_3 x_2)$$

is a multiplication operator. Hence $M_m \partial_3^m$ commutes with X_m , and

$$\begin{aligned} [X_m, [L^{[0]}, X_m]] &= [k_3 x_1 + h_3 x_2, k_3 \nabla_1 + h_3 \nabla_2] \partial_3^{2m} \\ &= -(k_3^2 + h_3^2) \partial_3^{2m} \\ &= -\mu \partial_3^{2m}. \end{aligned}$$

Since $\mu > 0$, we can derive iteratively $\partial_3^{2m} \in \widehat{E}$, which contradicts to the finite-dimensionality.

4 Proof of $k_1 = k_2 = h_2 = 0$

Proof. Suppose, to the contrary, that

$$M = \begin{pmatrix} k_1 & k_2 \\ k_2 & h_2 \end{pmatrix} \neq 0.$$

Since M is real symmetric, an orthogonal change of the visible variables diagonalizes it. After interchanging the two visible axes if necessary, and then translating x_1 , we may assume

$$k_2 = 0, \quad k := k_1 \neq 0, \quad k_0 = 0.$$

Write $h = h_2$.

After rechoosing the Poincaré representative, we have

$$f_1 = 0, \quad f_2 = b_0 x_1, \quad f_3 = F(x_1, x_2),$$

where

$$F(x_1, x_2) = \frac{k}{2} x_1^2 + \frac{h}{2} x_2^2 + h_0 x_2.$$

Thus

$$D_1 = \partial_1, \quad D_2 = \partial_2 - b_0 x_1, \quad D_3 = \partial_3 - F.$$

All coordinate changes and gauge conjugations used below are Lie-algebra isomorphisms and therefore preserve finite-dimensionality.

Step 1: the only possible x_3 -dependence of η .

In the present normal form,

$$H_0 = [L_0, D_1] = b_0 D_2 + k x_1 D_3 + \frac{1}{2} \eta_1,$$

and

$$H_1 = [L_0, D_2] = -b_0 D_1 + (h x_2 + h_0) D_3 + \frac{1}{2} \eta_2.$$

Direct normal ordering gives

$$\begin{aligned} [D_1, [D_1, H_0]] &= \frac{1}{2} \partial_1^3 (\eta - F^2), \\ [D_2, [D_1, H_0]] &= \frac{1}{2} \partial_1^2 \partial_2 (\eta - F^2), \\ [D_2, [D_2, H_0]] &= \frac{1}{2} \partial_1 \partial_2^2 (\eta - F^2), \\ [D_2, [D_2, H_1]] &= \frac{1}{2} \partial_2^3 (\eta - F^2). \end{aligned}$$

The four left-hand sides are multiplication operators in E . Since

$$E \cap C^\infty(\mathbb{R}^3) = \text{span}_{\mathbb{R}}\{1, x_1, x_2\},$$

they are independent of x_3 . Consequently, for arbitrary s, t , the function

$$\eta(x_1, x_2, s) - \eta(x_1, x_2, t)$$

is a polynomial of degree at most two in x_1, x_2 . Therefore, η can be written as

$$\eta = \eta_0(x_1, x_2) + (x_1, x_2)^T Q(x_3)(x_1, x_2) + \ell(x_3) \cdot (x_1, x_2) + e(x_3),$$

where $Q(x_3)$ is symmetric, $\ell(x_3) \in \mathbb{R}^2$, and η_0 is smooth.

Now $[D_2, H_0]$ is a function element, so the off-diagonal entry of Q is constant. Moreover,

$$h[D_1, H_0] - k[D_2, H_1]$$

is a function element because its D_3 -coefficient is zero. It follows that

$$hQ_{11} - kQ_{22}$$

is constant. Absorbing the constant quadratic terms into η_0 , we may therefore write

$$Q(x_3) = \varepsilon(x_3) \begin{pmatrix} k & 0 \\ 0 & h \end{pmatrix}$$

for some smooth function ε .

The genuine Lie element

$$T := \frac{1}{k} [D_1, H_0]$$

then has the form

$$T = D_3 + \varphi(x_1, x_2) + \varepsilon(x_3).$$

A further direct calculation gives

$$[T, H_0 - b_0 D_2] = \frac{1}{2} \ell'_1(x_3)$$

and

$$[T, H_1 + b_0 D_1] = \frac{1}{2} (\ell'_2(x_3) - 2h_0 \varepsilon'(x_3)).$$

Both sides are function elements depending only on x_3 , and hence are constants. Thus

$$\ell(x_3) = 2h_0 \varepsilon(x_3) \begin{pmatrix} 0 \\ 1 \end{pmatrix} + a + x_3 a'$$

for some constant vectors $a, a' \in \mathbb{R}^2$. We have proved

$$\eta = \eta_0(x_1, x_2) + 2\varepsilon(x_3)F(x_1, x_2) + e(x_3) + (a + x_3 a') \cdot y. \quad (4.1)$$

Step 2: a hidden gauge and polynomiality.

Choose a smooth univariate function Λ satisfying

$$\Lambda' = \varepsilon$$

and set

$$\tilde{E} = e^\Lambda E e^{-\Lambda}.$$

This conjugation fixes multiplication operators and sends

$$D_3 = \partial_3 - F \quad \text{to} \quad \partial_3 - (F + \varepsilon).$$

It also sends T to

$$\tilde{T} = \partial_3 + g(x_1, x_2)$$

for some smooth function g .

Set

$$r(x_3) := e(x_3) - \varepsilon(x_3)^2.$$

Expanding the conjugated L_0 and using (4.1), the terms $2\varepsilon F + \varepsilon^2$ cancel. Hence

$$\begin{aligned} \tilde{L}_0 = & \frac{1}{2}(\partial_1^2 + \partial_2^2 + \partial_3^2) - b_0 x_1 \partial_2 - (F + \varepsilon) \partial_3 \\ & + V_0(x_1, x_2) + x_3 V_1(x_1, x_2) - \frac{1}{2}(\varepsilon'(x_3) + r(x_3)), \end{aligned} \quad (4.2)$$

where the precise forms of the visible functions V_0, V_1 will not be needed.

The coefficient of ∂_3 in

$$[\tilde{L}_0, \tilde{T}]$$

is $\varepsilon'(x_3)$. If this commutator has order one, the theorem on polynomial highest-order coefficients, transported through the gauge conjugation, shows that ε' is a polynomial. If it has order zero, then $\varepsilon' = 0$. Thus

$$\varepsilon \in \mathbb{R}[x_3].$$

Let $d = \deg \varepsilon$, with $d = 0$ when $\varepsilon = 0$, and put

$$m = \max\{2, d\}, \quad c = \varepsilon^{(m)}.$$

Then c is a real constant. Since

$$\tilde{T} = \partial_3 + g(x_1, x_2), \quad [\partial_3, g] = 0,$$

and every commutator with the multiplication operator g lowers differential order, direct expansion gives

$$\text{ad}_{\tilde{T}}^m(\tilde{L}_0) + c\tilde{T}$$

as a multiplication operator. Its only x_3 -dependent part is

$$-\frac{1}{2}r^{(m)}(x_3).$$

It belongs to

$$\tilde{E} \cap C^\infty(\mathbb{R}^3) = \text{span}_{\mathbb{R}}\{1, x_1, x_2\},$$

so $r^{(m)}$ is constant. Therefore

$$r \in \mathbb{R}[x_3], \quad \deg r \leq \max\{2, \deg \varepsilon\}.$$

Step 3: a weighted Lie ladder.

Every coefficient in (4.2) is now polynomial in x_3 . For a normally ordered monomial define

$$v\left(a(x_1, x_2)x_3^p \partial_1^{q_1} \partial_2^{q_2} \partial_3^{q_3}\right) := p + q_1 + q_2 + 2q_3.$$

The visible coefficient $a(x_1, x_2)$ is not counted. The weight of an operator is the largest weight of its monomials.

A direct application of the Leibniz rule gives the only estimates needed:

- the constant second-order part and $-F\partial_3$ raise the weight by at most 1;
- $-c_s x_3^s \partial_3$ raises it by at most $s - 1$;
- multiplication by a polynomial of degree D raises it by at most $D - 3$;
- all remaining terms in (4.2) raise it by at most 0.

Consider

$$S_n := (\text{ad}_{\tilde{L}_0})^n(\partial_1) \in \tilde{E}$$

(i) If $\deg \varepsilon \geq 3$:

Write

$$\varepsilon(x_3) = ax_3^d + \text{lower powers}, \quad a \neq 0.$$

Since $\deg r \leq d$, the unique term that can raise the weight by $d - 1$ is

$$-ax_3^d \partial_3.$$

Moreover,

$$S_1 = kx_1 \partial_3 + \text{terms of weight at most 1}$$

and

$$[-ax_3^d \partial_3, x_1 x_3^q \partial_3] = a(d - q)x_1 x_3^{q+d-1} \partial_3.$$

It follows inductively that the highest-weight term of S_{n+1} is

$$ka^n \prod_{j=0}^{n-1} (d - j(d - 1)) x_1 x_3^{n(d-1)} \partial_3.$$

with $d - j(d - 1) \neq 0$ for $d \geq 3$ as an integer. This derives contradiction.

(ii): $\deg \varepsilon \leq 2$, including $\varepsilon = 0$.

Let a be the coefficient of x_3^2 in ε , allowing $a = 0$. Every bracket with $\tilde{L}_0$ now raises the weight by at most 1.

Let ξ, ζ be the graded symbols of ∂_1, ∂_3 . Starting from ξ , the part that raises the weight by exactly 1 is the derivation

$$\mathcal{D} = \xi \partial_{x_1} + (\zeta - ax_3^2) \partial_{x_3} + kx_1 \zeta \partial_\xi + 2ax_3 \zeta \partial_\zeta.$$

The terms containing h and h_0 act only on the (x_2, ∂_2) -variables and therefore never enter $\mathcal{D}^n \xi$.

We claim that

$$\mathcal{D}^n \xi \neq 0 \quad (n \geq 0).$$

Otherwise, suppose that $\mathcal{D}^N \xi = 0$. Along every local integral curve of $\mathcal{D}$, both $\xi(\tau)$ and $x_1(\tau)$ would then be polynomials. The relevant equations are

$$\dot{x}_1 = \xi, \quad \dot{\xi} = kx_1 \zeta, \quad \dot{x}_3 = \zeta - ax_3^2, \quad \dot{\zeta} = 2ax_3 \zeta.$$

Choose

$$x_3(0) = 0, \quad \zeta(0) = 1.$$

For this fixed hidden trajectory, the two initial conditions

$$(x_1(0), \xi(0)) = (1, 0), \quad (x_1(0), \xi(0)) = (0, 1)$$

give two polynomial solutions p, q of

$$u'' = k\zeta(\tau)u.$$

Their Wronskian is the constant 1.

Two polynomials with a nonzero constant Wronskian span a nonzero constant polynomial: after making their degrees unequal by a constant change of basis, the degree of their Wronskian is the sum of their degrees minus one. Hence those degrees must be 0 and 1. The equation therefore has a nonzero constant solution u , which would imply

$$0 = k\zeta(\tau)u.$$

This contradicts $k \neq 0$ and $\zeta(0) = 1$. Thus

$$\mathcal{D}^n \xi \neq 0 \quad (n \geq 0).$$

Consequently, the highest-weight term of S_n is $\mathcal{D}^n \xi$, and

$$\nu(S_n) = n + 1 \longrightarrow \infty.$$

Therefore $M = 0$. Returning to the original visible coordinates gives

$$k_1 = k_2 = h_2 = 0.$$

The proof divides only by $k \neq 0$, so it includes the rank-one case $h = 0$ and allows arbitrary h_0 . □

5 Conclusion

Every unbounded sequence in the proof consists of actual Lie words. The proof for the visible entry uses only the function-element space and the affine Wong structure. In Sector I, the spectral projections are introduced only after Cayley–Hamilton and the global smooth Euler equation have produced a finite spectrum, and they are then realized by the Chinese remainder theorem. Sector II is treated inside $C^\infty(\mathbb{R}^2)[x_3]\langle \partial_1, \partial_2, \partial_3 \rangle$. Thus we never treat the Lie algebra as a module over functions, nor do we use analytic continuation, maximal rank, or an a priori polynomial η .

Jiao’s Theorem 5.4 also derives, from the structure of η , polynomiality of the pre-gauge term r . The smooth Euler lemma in the present paper additionally controls the post-gauge zeroth-order slot U , so the final argument remains strictly within the smooth assumptions of the original problem.

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